Evaluating English-to-Romanized Indic Generation with an Open-Weight Language Model

Anonymous authors
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Abstract

Romanized text is a common interface for speakers of Indic languages, yet general-purpose language models are rarely evaluated on long-form generation in this representation. We present a preliminary evaluation of English-to- Romanized Hindi, Telugu, and Kannada generation using an open-weight Qwen3.5-35B model served locally through Ollama. We generate romanized versions of 142 English query–response pairs and evaluate semantic preservation with raw multilingual BERTScore F1 and BGE-M3 embedding cosine similarity. Hindi obtains the highest query and response cosine similarity, at 0.735 and 0.787, respectively. English-copy, blank, and malformed failures receive zero. Kannada obtains a response cosine similarity of 0.455 after all 142 stored rows enter the mean. These results illustrate how failure-aware scoring changes model assessment and motivate reporting failure policy alongside semantic metrics.

1 Introduction

Many speakers of South Asian languages communicate digitally in a native language written with Latin characters. This practice is widespread but non-standard: spelling varies across speakers, platforms, and contexts, and English words are often mixed into otherwise Indic-language utterances. Datasets such as Dakshina document the importance of Latin-script South Asian language processing Roark et al. (2020), while Aksharantar demonstrates the scale required for robust Indic transliteration systems Madhani et al. (2023).

Romanized generation presents a compound challenge for a general-purpose language model. The model must preserve the source meaning, produce natural target-language phrasing, and express that phrasing in a variable Latin-script form. A model can fail at any stage by copying English, switching to native script, omitting content, returning malformed structured output, or entering a repetition loop. A semantic score calculated only over successful generations can consequently overstate system quality.

We study this problem through a controlled English-to-Romanized Indic generation experiment. A locally served Qwen3.5-35B model converts paired English queries and responses into Romanized Hindi, Telugu, and Kannada. We evaluate the outputs with multilingual BERTScore and BGE-M3 cosine similarity, while separately tracking the number of scorable outputs.

Our preliminary contributions are:

- an end-to-end local pipeline for paired query–response romanization in three Indic languages;
- a comparison of token-alignment and embedding-based semantic metrics for English-to-romanized generation; and
- evidence that metric averages must be reported together with output coverage, because missing generations materially change interpretation.

	Query		Response		Coverage	
Language	BERT-F1	Cosine	BERT-F1	Cosine	Query	Resp.
Hindi	0.622	0.735	0.639	0.787	142	142
Telugu	0.460	0.522	0.459	0.527	142	142
Kannada	0.514	0.577	0.412	0.455	142	142

Table 1: Failure-aware mean automatic scores. All 142 samples enter each mean; BERT-F1 is raw multilingual BERTScore F1 and Cosine uses BGE-M3 embeddings.

2 Method

2.1 Data

We use 142 English query–response pairs derived from a processed subset of the OpenAssistant dataset. Each record contains a message identifier, an English query, and an English response. The same records are used for Hindi, Telugu, and Kannada generation. This setup tests translation and romanization rather than understanding of naturally occurring Romanized Indic prompts.

2.2 Generation

We serve the quantized qwen3.5:35b model through Ollama on an NVIDIA RTX PRO 6000 Blackwell GPU. The model receives the English query and response together in one request and is instructed to return a JSON object containing romanized_query and romanized_response. Joint generation preserves the relationship between both fields and avoids the context loss observed in separate-call pilot experiments.

We disable model thinking, use temperature zero, and permit up to 4096 generated tokens. Requests time out after 300 seconds. Malformed or partial outputs are retained when recoverable; failed fields remain blank. Kannada uses a stricter language-specific prompt that explicitly prohibits English and Kannada script in the final output.

2.3 Evaluation

For each non-empty generated field, we compare the English source against its romanized counterpart with two automatic metrics.

Raw BERTScore F1. We use bert-base-multilingual-cased with baseline rescaling disabled Zhang et al. (2020). The resulting value is the raw BERTScore F1. Because the source and output differ in both language and surface form, this score is treated as a secondary token-alignment signal.

BGE-M3 cosine similarity. We embed the English and romanized texts with BGE-M3 and calculate cosine similarity Chen et al. (2024). This cross-lingual embedding score is our primary automatic semantic measure.

Coverage. We retain the first Qwen3.5-35B generation for each message identifier and discard duplicate reruns. If either generated field is identical to its whitespace-normalized English source, we assign zero to all query and response metrics for that row. Blank or malformed fields receive zero for their corresponding metrics. Thus all 142 samples enter each aggregate mean.

3 Results

Table 1 shows the automatic evaluation results. Rows containing an output identical to its English source receive zero for all metrics. Blank and malformed fields also receive zero. We

retain the first Qwen3.5-35B result for each message identifier and remove duplicate reruns. Hindi has the highest query cosine similarity (0.735) and response cosine similarity (0.787).

Kannada illustrates the importance of joint quality and reliability reporting. Its response BERTScore F1 is 0.412 and its response cosine similarity is 0.455 when all 142 rows, including failures scored as zero, contribute to the mean.

Telugu uses the same 142-sample denominator as the other languages. English copies, malformed JSON, truncated output, and repetition failures contribute zero. Consequently, Telugu's conditional semantic scores should not be interpreted as a 142-record success rate.

4 Discussion

Semantic quality and reliability are different axes. Embedding similarity captures whether a completed output remains semantically close to its source. Our failure-aware means additionally penalize missing, unusable, and English-copy outputs with zero. Hindi's stronger aggregate scores therefore reflect both semantic preservation and more reliable completion.

Structured generation is part of the task. Several failures arise before linguistic quality can be evaluated. Long responses may exhaust the generation budget before the second JSON field, while repetition loops consume tokens without completing the requested structure. Evaluations that silently remove such records measure conditional quality rather than operational utility.

Automatic metrics are insufficient for naturalness. Neither BERTScore nor BGE-M3 evaluates whether a romanization sounds natural to a Hindi, Telugu, or Kannada speaker. High semantic similarity can coexist with awkward grammar, inconsistent spelling, inappropriate register, or excessive English borrowing. Native-speaker judgments are required to establish practical quality.

Implications for multilingual evaluation. Romanized generation should be evaluated with a metric bundle: semantic preservation, completion rate, script compliance, repetition rate, and native-speaker acceptability. The failure-scoring policy must be stated explicitly because excluding unsuccessful outputs can substantially inflate aggregate quality.

5 Limitations

This study evaluates one quantized model and one prompting pipeline, so the results do not establish a general ranking of model families. The source data is English rather than naturally occurring Romanized Indic input, and the experiment therefore measures controlled translation and romanization.

The current evaluation contains no native-speaker ratings. Automatic cross-lingual similarity cannot establish fluency, register appropriateness, cultural fit, or standardized transliteration quality. Raw multilingual BERTScore is particularly difficult to interpret across English and Romanized Indic text because its token alignment is sensitive to surface vocabulary.

English-copy outputs are treated as failures and assigned zero across the row. Blank or malformed fields also receive zero for the affected metric. This failure-aware policy is intentionally strict and may conflate infrastructure failures with linguistic failures. For duplicate Telugu reruns, we retain the first generation; a different selection rule could produce slightly different aggregates. All experiments should still be rerun from a clean manifest before submission.

Finally, romanization has no single universally correct spelling. Future work should evaluate multiple acceptable forms and recruit native speakers from different regions and demographic backgrounds.

6 Conclusion

We presented a preliminary evaluation of Qwen3.5-35B for paired English-to-Romanized Hindi, Telugu, and Kannada generation. Under a failure-aware policy that assigns zero to English copies and missing or unusable fields, Hindi achieves the strongest aggregate semantic scores, followed by Telugu and Kannada. Future evaluation should combine automatic semantic metrics with completion and format diagnostics, repetition detection, and native-speaker judgments.

References

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Reproducibility Statement

The generation and evaluation scripts, model identifiers, prompts, decoding parameters, and aggregate results are maintained in the accompanying anonymous artifact. Before submission, the artifact will include a clean input manifest, software versions, hardware details, and commands required to reproduce every table.

Use of Language Models

Language models were used in the research workflow for data generation, evaluation, code assistance, and drafting assistance. The authors reviewed all generated text, code, and claims and remain responsible for the contents of the paper.